Air and water poisoning

Any kind of pollution threatens biodiversity, but air and water pollution are particularly harmful. Burning fossil fuels, for example, releases the waste gases sulfur dioxide and nitrogen oxide into the air; these return as acid rain, causing water and soil acidification and affecting ecosystem health and biodiversity. Ozone emissions at ground level can also damage cell membranes on plants, curbing their growth and development. Water pollution is mainly due to sewage or chemicals absorbed into water as it flows off agricultural land, which reduces oxygen levels in water. This makes survival more difficult for some species, especially when combined with water temperatures that have risen due to climate change. Streams used by spawning fish, for example, can be made uninhabitable by pollution. Water pollution caused by agricultural fertilizers can lead to excessive algal blooms or plant growth (eutrophication). This can have multiple detrimental effects for species inhabiting the water, such as blocking the light needed to hunt prey, or making it unsafe for terrestrial species to drink.

Some organisms can absorb a substance, such as an agricultural chemical, more quickly than they can excrete it, in a process known as bioaccumulation. Initial, low concentrations of chemicals may not be a problem. However, as those chemicals accumulate through the food chain—from phytoplankton to fish to mammal, for example—they can reach levels that cause birth defects and disrupt hormone levels and immune systems.

Rapid human population growth further damages the environment. It has risen from less than 1 billion in 1800 to more than 7 billion, and is expected to reach nearly 10 billion by 2050. As the population grows, so do other threats to biodiversity: increasing numbers of invasive species are spread through trade and travel; urban development and resource extraction destroy habitats; more pollution is created; and land is overharvested. The